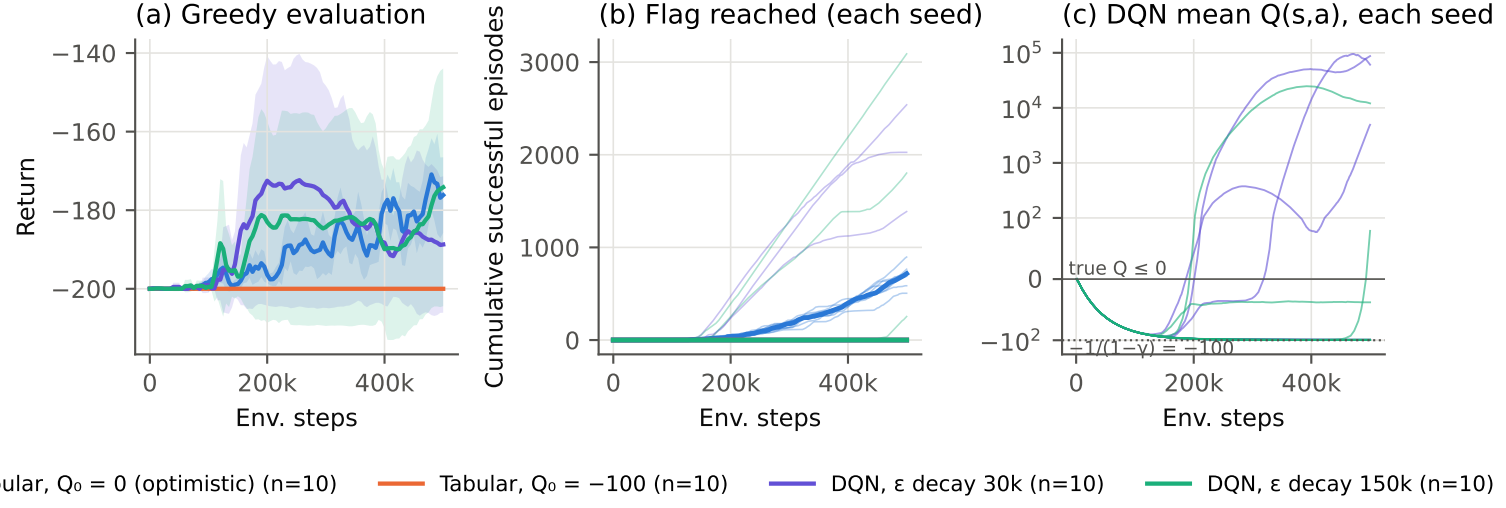
(a) Greedy evaluation
Return
−140
−160
−180
−200
0
200k
400k
Env. steps
(b) Flag reached (each seed)
Cumulative successful episodes
3000
2000
1000
0
0
200k
400k
Env. steps
(c) DQN mean Q(s,a), each seed
10^5
10^4
10^3
10^2
0
-10^2
true Q ≤ 0
=1/(1−γ) = −100
0
200k
400k
Env. steps
Tabular, Q₀ = 0 (optimistic) (n=10)
Tabular, Q₀ = −100 (n=10)
DQN, ε decay 30k (n=10)
DQN, ε decay 150k (n=10)